

Operationalizing Self-Perceptions of Aging Using Reflection Probe and SPA Design Cards with HCI Practitioners

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Abstract

Self-Perceptions of Aging (SPA) are associated with older adults' emotional, cognitive, and physical wellbeing and can be influenced through targeted interventions, yet SPA is rarely operationalized as concrete design material in HCI. This paper presents a material-driven method for using SPA as a lens for wellbeing-oriented design with HCI practitioners who are not themselves older adults. Building on formative studies with 24 older adults, we developed three interconnected SPA materials: an AI-mediated self-reflection tool grounded in Indian scripture texts, empathic reflection cards combining older adults' quotes with emotion labels and corresponding guidance, and SPA design aspiration cards distilled from a systematic review of SPA intervention effects. These materials were used with 10 HCI practitioners in a structured session progressing from personal reflection to interpreting older adults' situations. Analysis of discussions and follow-up feedback shows how SPA can be brought more concretely into HCI design practice.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Ethnographic studies**; *HCI theory, concepts and models*.

Keywords

Older adults; Aging technologies; Cross-cultural HCI; Ethnography; Qualitative methods; Semi-structured interviews; Empirical study; Technology design implications

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1 Introduction

Self-Perceptions of Aging (SPA) describe how individuals understand and evaluate their own aging, including perceptions of their capabilities, social roles, and future expectations [3]. Prior work shows that more positive SPA are associated with better emotional wellbeing, cognitive functioning, physical health, and longevity, whereas negative SPA are linked to higher anxiety and depression, poorer functional health, and lower quality of life [9, 14, 15]. Importantly, SPA are not fixed. A growing body of evidence demonstrates that interventions such as psychoeducation, stereotype-challenging activities, volunteering, leisure engagement, and life-review programmes can shift SPA in more positive directions [2, 12, 17].

Despite this evidence, SPA is rarely operationalized as concrete design material in HCI. Age is often treated as a demographic or biomedical variable, rather than a multidimensional (emotional, cognitive, physical, social) and modifiable construct, shiftable via interventions like exergames [4, 12], that can inform design decisions [8]. This gap persists even as HCI research increasingly calls for approaches that move beyond chronological age and translate insights from aging research into tools that support reflective and context-sensitive design practice [8, 11]. As a result, HCI practitioners lack practical means to connect SPA theory, empirical findings on aging, and everyday ideation work when designing wellbeing-oriented technologies.

Card-based toolkits are established in HCI for structured ideation and reflection [1, 16]. Building on that work, this paper introduces a material-driven method for operationalizing SPA in HCI design practice, part of a larger Research through Design program [6, 7]. Drawing on formative studies with 24 older adults in India, we developed three interconnected SPA materials: (1) an AI-mediated SPA self-reflection tool grounded in Indian scriptures such as the Bhagavad Gita and selected Upanishads; (2) empathic reflection cards that combine older adults' quotes, derived emotion labels, and



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tool-generated guidance; and (3) SPA design aspiration cards distilled from a systematic review of SPA intervention effects [14, 17]. We used these materials in a structured session with HCI practitioners that progressed from personal reflection, to interpretation of older adults' situations, to the generation of wellbeing-oriented design concepts.

We ask: RQ1) How can SPA be translated into reflection probes and design cards that are usable by HCI practitioners? RQ2) How do practitioners use these SPA materials when interpreting older adults' situations and generating wellbeing-oriented design ideas? Our analysis shows that the materials supported a shift from generalized assumptions about "older people" toward SPA-dimension-specific interpretations and early design concepts, illustrating how SPA can function as a concrete design lens in HCI practice.

2 Method

This study follows a material-driven research through design approach. We first developed SPA materials grounded in formative work and intervention literature, and then examined how HCI practitioners engaged with these materials in a structured workshop.

2.1 Background and Materials

We conducted formative studies with 24 older adults in India across private homes, assisted living facilities, and old age homes to examine lived experiences related to nine aspects of Self-Perceptions of Aging (SPA). These studies served as the empirical basis for developing the SPA self-reflection tool and the empathic reflection cards. Semi-structured interviews lasting 60–70 minutes were transcribed and analysed using qualitative content analysis [5], supported by AI-assisted coding in ATLAS.ti to identify recurring emotional expressions. From this analysis, we selected 25 frequently occurring emotions (e.g., frustration, nostalgia, concern, happiness, acceptance) and paired each with an illustrative quote. These quotes were then input into the SPA self-reflection tool to generate guidance texts, resulting in empathic reflection cards that combine an older adult's quote, associated emotion labels, and corresponding AI-generated guidance.

SPA design aspiration cards were created based on a review of intervention studies that examine how SPA can be influenced [17]. Each of the ten cards summarizes an evidence-based direction for shaping SPA, such as correcting misconceptions about aging, emphasizing age-related gains and positive self-image, combining leisure with structured activity, or using positive feedback and social comparison to support self-efficacy. These cards exhibit minor overlaps that reflect natural synergies across SPA intervention strategies in the literature [14, 17]. They were designed to translate findings from psychological and gerontological research into concise prompts that can support ideation in design practice.

The AI self-reflection tool serves dual roles as a prototype intervention for older adults (under evaluation) and experiential design material for practitioners to critique its reflective capabilities. The tool draws on Indian scriptures as its cultural corpus, primarily the Bhagavad Gita and selected Upanishads. The SPA self-reflection tool draws on Indian scriptures as its cultural corpus, primarily the Bhagavad Gita and selected Upanishads. AI works as a parser and

curator that analyzes users' reflections to detect emotions and situations, then retrieves and contextualizes relevant scriptures from a curated database to provide personalized guidance. These nine SPA aspects (social presence, self-efficacy, activities, aging process, life satisfaction, feeling valued, agency, age association, and emotions) were identified in prior work [7] reviewing standard SPA measures (ASD, PGCM, APQ, AAQ-24, ATOA). Users select one of the nine SPA aspects, enter a brief reflection, and receive guidance that integrates empathetic acknowledgement, a short interpretation of a relevant scripture, and a small set of concrete suggestions. Guidance is generated through a constrained language-model interface that retrieves and recombines verses and explanations from a curated database. The database contains 700 clean Bhagavad Gita verses and 300 manually selected Upanishad verses (1000 total), chosen for relevance to aging, emotional well-being, and self-reflection themes. Together, the self-reflection tool, empathic reflection cards, and SPA design aspiration cards were used as materials in conducting a workshop with HCI practitioners. As a set, these materials translate empirical insights from older adults and evidence from SPA intervention literature into resources that can support a broader SPA-oriented design process.

2.2 Workshop with HCI Practitioners

We ran a 90-minute workshop with HCI practitioners at a regional HCI workshop to explore how the SPA materials (Figure 1a, Figure 1b, Figure 1c) could support wellbeing-oriented design. Participants were introduced to the three material types and invited to use them to generate SPA-oriented concepts for older adults in India.

The workshop had four phases: (1) a brief orientation on aging in India, SPA, and the role of cultural and spiritual frameworks; (2) individual use of the self-reflection tool followed by whole-group reactions to AI based guidance; (3) group work with empathic reflection cards and SPA aspiration cards to discuss older adults' situations and possible directions; and (4) small-team ideation where participants combined cards to sketch SPA-oriented wellbeing interventions, including services, spaces, and technologies.

Ten HCI practitioners participated (six female, four male), including researchers, interaction designers, and professional practitioners, mostly aged 21–35, with one aged 55. All were conference attendees, fluent in English, and varied in prior aging-related experience. Participants were predominantly younger than the target population and from urban, middle or upper-middle class backgrounds, so their perspectives are best understood as practitioner views on how SPA materials can support design work.

Data included audio recordings of workshop discussions, photographs of annotated materials, collected design artefacts (sketches, concept maps), and post-workshop feedback forms. Post-workshop, participants received digital materials and provided optional one-week feedback via email (6/10 responded). Recordings were transcribed and translated where needed. Inductive qualitative content analysis identified segments where participants reflected on their tool experience, interpreted older adults' situations, reacted to AI curated guidance text, discussed SPA aspects or aspirations, or explained how they developed concepts. Related codes were grouped into patterns describing how participants talked about aging and

SPA, how they used the materials during ideation, and where they raised concerns or questions, which were then synthesized into themes about how SPA functioned as a design resource. The analysis focused on understanding how SPA was taken up as a design lens rather than on evaluating technical quality or feasibility of specific concepts.

2.2.1 Positionality. The first author is a PhD researcher in interaction design and aging studies with experience conducting formative work with older adults in India; this provides familiarity with local languages, spiritual practices, and aging contexts but does not cover the full diversity of region, class, caste, or religion. The workshop and analysis centre the perspectives of HCI practitioners at a conference and do not substitute for direct engagement with older adults when evaluating any proposed interventions.

3 Findings

Analysis of workshop discussions, design artefacts, and follow-up feedback shows how the SPA materials functioned as design resources and how they shaped participants' talk and concepts around aging and SPA. Below, the findings are organised around personal use of the self-reflection tool, work with older adults' reflection cards, use of SPA design aspiration cards in ideation, and critical perspectives on SPA and AI-mediated tools.

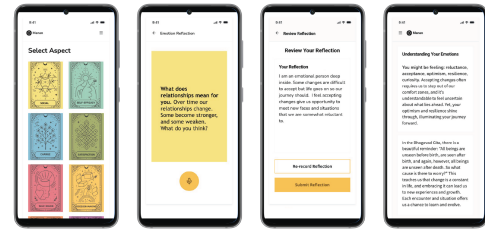
3.1 Personal reflection with the SPA tool

When participants used the SPA self-reflection tool for their own reflections, they focused on its emotional tone, cultural resonance, and limits. Several described the guidance as gentle and reassuring, "supportive and calming," and recognised familiar concepts such as *seva* (service) that felt like "something we subconsciously know." Repeated use also made the tool's constraints visible. Some participants noticed similar-sounding guidance across different reflections and compared it to horoscopes that "all sound similar," while others pointed out keyword-driven responses, such as the system reacting to "reluctance" even when someone wrote "I am not reluctant to change."

These observations led participants to question how older adults might interpret such patterns and what expectations the system could safely set. They also raised practical concerns, including the need for voice input saying "my parents will not type." They also highlighted a need for clearer prompting so users know what to write about, and more transparency about "who is speaking" in the guidance, whether it should be perceived as a friend, expert, or divine voice. These concerns later shaped how they engaged with the reflection and aspiration cards.

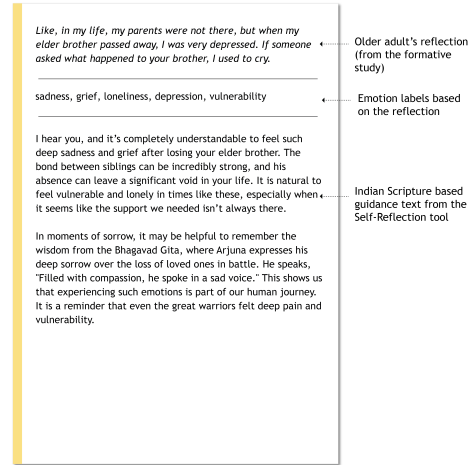
3.2 Working with older adults' reflection cards

Moving from their own reflections to older adults' reflection cards produced a noticeable shift in how participants talked about aging. Early in the workshop, older adults were often discussed via broad cultural narratives and stereotypes (for example, "old people" being self-absorbed or gendered assumptions about post-retirement happiness). As they read and discussed the cards where each card combined a short quote, emotion labels, and AI generated guidance, participants' insights became more specific and situational. Participants discussed concrete stories, such as a woman whose sewing



A) SPA Self-Reflection tool where users select a SPA theme card, enter a brief reflection, and receive AI-mediated guidance grounded in Indian scriptures.

(a) Self-reflection tool



B) Empathic reflection card combining an older adult's quote from formative studies, derived emotion labels, and the corresponding guidance text.

(b) Empathic reflection card



C) SPA design aspiration cards distilled from a systematic review of SPA intervention effects, each summarizing an evidence-based direction for influencing SPA.

(c) Design aspiration cards

Figure 1: SPA materials used in the workshop. (A) The SPA self-reflection tool, where users select an SPA theme, enter a brief reflection, and receive AI-mediated guidance grounded in Indian scriptures. (B) An empathic reflection card combining an older adult's quote from formative studies, derived emotion labels, and the corresponding guidance text generated by the tool. (C) SPA design aspiration cards distilled from a systematic review of SPA intervention effects, each summarizing an evidence-based direction for influencing self-perceptions of aging.

machine was taken without asking and who felt taken for granted, a man who was isolated and "starts crying when someone asks about

him,” or a person who feared falling because of responsibilities “if something happens.”

By focusing on these narratives, participants began to identify dilemmas like being taken for granted, managing grief alone, or fearing dependency, rather than treating “older adults” as a homogeneous group. One-week feedback echoed this shift, with a participant writing that people who are aging “have some shared experiences but are unique human beings with their own preferences and perspectives,” and another describing SPA as “new to me” and “more contextual and nuanced based on target user group.” Several explicitly highlighted “hearing actual words from people in the target group” and “selecting the prompts and reading the stories on the cards” as the most engaging parts of the session.

3.3 Using SPA design aspiration cards in ideation

In the final phase, teams combined reflection cards with SPA design aspiration cards to frame opportunities and sketch interventions. Each team selected stories that resonated with them and then chose aspirations such as “using positive feedback and social comparison to build self-efficacy” or “emphasising age-related gains and positive self-image” to guide their thinking, treating these aspirations as concrete levers for changing SPA rather than as generic slogans. SPA vocabulary helped teams analyse situations and define what their concepts should support; for instance, one group linked a story about feeling useless after retirement to self-value and self-efficacy, while another interpreted the sewing-machine case through agency and decision-making.

Resulting concepts were varied but consistently grounded in SPA-related concerns and Indian social and spiritual contexts. Examples included a dual-mode support system where older adults could either speak freely or request concrete suggestions, combined with nostalgia-based experiences such as revisiting familiar places; a volunteering-oriented idea where people “be the support you do not have” to counter loneliness; identity-affirming community activities rather than a single app; and combinations of home environment changes, intergenerational support, and participation in satsang or group activities to support confidence and belonging. Participants later described the SPA design cards and ideation exercise as helpful for “discovering gaps and potential solutions.”

3.4 Critical perspectives on SPA and AI-mediated tools

Participants also raised critical questions about the role of AI and spiritual framing in SPA-oriented interventions. Some noted that many older adults already have rich spiritual practices and communities, such as regular listening to spiritual discourses or online satsang groups formed during COVID that function as “therapy,” and wondered how an AI-mediated tool would add value for them. Others pointed to accessibility and possible over-reliance. They raised concerns about older adults who are “very averse to reading and writing,” gave the example of a middle-aged woman who talks with an AI system every night while eating dinner as a potential case of dependency, and observed that some grandparents “do not want to answer what you ask, they just want to speak what they want to say.” These concerns reflect broader HCI discussions

about potential over-reliance on AI companions among older adults, highlighting the need for hybrid human-AI approaches (e.g., human oversight of AI companions, community review systems) that complement existing social and spiritual practices [10, 13].

These reflections highlight tensions that SPA-oriented, AI-supported interventions must address. Participants questioned how to complement rather than duplicate existing social and spiritual practices, how to avoid fostering unhealthy dependence on AI, and how to tune interaction styles to older adults’ preferred ways of expressing themselves. In the workshop, the SPA materials did not resolve these tensions, but they provided a concrete basis for articulating them and raising questions about the appropriateness of technological interventions and their boundaries.

4 Discussion

The workshop illustrates how SPA-oriented materials can operationalize a gerontological construct for HCI practice and begin to shift how practitioners talk and design around aging. Personal use of the Self-Reflection tool gave participants an experiential grounding in the tone, cultural resonance, and limits of AI- and Indian scriptures-based AI-curated guidance, making the output from the tool something that could be felt and critiqued. Reflection cards, built from older adults’ quotes, emotion labels, and AI-curated guidance, made SPA-related concerns tangible and particular, supporting a move from category-level talk about aging to situation-specific readings of self-doubt, grief, fear, and purpose. SPA design aspiration cards then bridged these stories and possible interventions by condensing intervention findings into short aspirations that helped participants articulate aims such as strengthening self-value, challenging misconceptions, or supporting agency.

The sequence of personal reflection, empathic engagement, and aspiration-driven ideation suggests a more general pipeline for translating constructs like SPA into design methods. This workshop with HCI practitioners complements ongoing co-design with older adults that generated the SPA materials [7], serving as an intermediate step within a larger Research through Design program exploring technology-mediated SPA interventions. While direct participatory design with older adults remains essential for design and development of tech-mediated interventions, these materials enable practitioners to engage with SPA concepts asynchronously and scalably before co-design sessions. Pairing lived narratives with construct-specific aspects and aspirations helped participants talk about emotions as well as about dimensions such as self-value or agency, and culturally grounded materials allowed the session to focus on aging experiences rather than basic explanation of context. At the same time, participants’ comments about AI, spirituality, and communication styles underline that operationalizing constructs like SPA cannot be separated from questions of power, access, and fit with existing practices, indicating that critical questioning should be built into such methods and implications of guardrails for such interventions.

5 Limitations

This study captures a single 90-minute workshop, and participants themselves noted time and flow issues. The findings therefore reflect an initial encounter with SPA-oriented materials rather than

longer-term integration into practice. The participant group was small and demographically narrow, consisting of ten younger, urban HCI practitioners at a regional conference. As a result, their interpretations cannot be taken as representative of designers in other regions, of older adults, or of caregivers. The materials are grounded in Indian contexts and Indian scriptures, and the study examines how practitioners engage with SPA as a design lens but does not test whether resulting concepts or implementations improve SPA or wellbeing for older adults.

6 Conclusion

This work presented a material-driven method for operationalizing Self-Perceptions of Aging in HCI design practice through an AI-mediated self-reflection tool, empathic reflection cards built from older adults' narratives, and SPA design aspiration cards distilled from intervention literature, used together in a workshop with HCI practitioners. The study shows that such materials can help practitioners move from generalized talk about aging to SPA-dimension-specific readings of concrete situations and can provide vocabulary and direction for framing wellbeing interventions in terms of SPA rather than only features or functions. While limited in scope and duration, the workshop points toward a broader approach for bringing constructs from gerontology into HCI by building construct-specific materials from empirical data and intervention evidence and using them in structured yet open design activities. Future work should involve older adults in evaluating SPA-oriented materials and interventions, adapt the material pipeline to other constructs and cultural contexts, and examine how repeated use of such materials might influence design practice over time. This workshop is one step in a larger research program focused on technology-mediated interventions to enhance SPA among older adults. Here, practitioner perspectives complement formative work with older adults and will inform subsequent iterations and deployments of SPA-oriented tools.

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References

- [1] Clara-Maria Barth, Anton Fedosov, Jürgen Bernard, Gabriela Morgenshtern, and Elaine M Huang. 2025. Incident Cards: A Tool for Documenting and Reviewing Experiences with Chronic Conditions. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [2] Ann-Kristin Beyer, Julia K Wolff, Ellen Freiburger, and Susanne Wurm. 2019. Are self-perceptions of ageing modifiable? Examination of an exercise programme with vs. without a self-perceptions of ageing-intervention for older adults. *Psychology & Health* 34, 6 (2019), 661–676.
- [3] Manfred Diehl and Hans-Werner Wahl. 2024. Self-perceptions of aging: A conceptual and empirical overview. *Current opinion in psychology* 55 (2024), 101741.
- [4] Yumei Dong and Hua Dong. 2023. Design empowering active aging: a resource-based design toolkit. *International Journal of Human-Computer Interaction* 39, 3 (2023), 601–611.
- [5] Hsiu-Fang Hsieh and Sarah E Shannon. 2005. Three approaches to qualitative content analysis. *Qualitative health research* 15, 9 (2005), 1277–1288.
- [6] Neeta M Khanuja. 2025. Designing Technological Interventions to Enhance Self-Perceptions of Aging (SPA) and Well-Being of Older Adults in India. In *Companion Publication of the 2025 ACM Designing Interactive Systems Conference*. 111–115.
- [7] Neeta M Khanuja, Valentina Nisi, and Jodi Forlizzi. 2025. Designing Aging Reflection Probes to Elicit Self-Perception of Aging (SPA) Beliefs of Older Adults in India. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 3735430. <https://doi.org/10.1145/3715336.3735430>
- [8] Bran Knowles, Jasmine Fledderjohann, Aneesha Singh, Richard Harper, Julia McDowell, Judith Tsouvalis, Alice Ashcroft, Yvonne Rogers, Ewan Soubutts, Andrew Steptoe, et al. 2025. Not just a number: a multidimensional approach to ageing in HCI. *Interacting with Computers* (2025), iwaf050.
- [9] Becca R Levy, Martin D Slade, Suzanne R Kunkel, and Stanislav V Kasl. 2002. Longevity increased by positive self-perceptions of aging. *Journal of personality and social psychology* 83, 2 (2002), 261.
- [10] Haoyan Li and Michael Chau. [n. d.]. Human-AI collaboration in content moderation: the effects of information cues and time constraints. *Journal of Management Information Systems* ([n. d.]).
- [11] Ittay Mannheim, Eveline JM Wouters, Hanna Köttl, Leonieke C Van Boekel, Rens Brankaert, and Yvonne Van Zaanen. 2023. Ageism in the discourse and practice of designing digital technology for older persons: A scoping review. *The Gerontologist* 63, 7 (2023), 1188–1200.
- [12] Reza Hadi Mogavi, Joseph Son, Sean Yang, Dakuo M Wang, Lucas Choong, Ali Alhilal, Pan Yu Zhou, Pan Hui, and Lennart E Nacke. 2024. The Jade Gateway to Exergaming: How Socio-Cultural Factors Shape Exergaming Among East Asian Older Adults. *Proc. ACM Hum.-Comput. Interact.* 8, CHI PLAY (2024), 341:1–341:34. <https://doi.org/10.1145/3677106>
- [13] Morteza Moradi, Mohammad Moradi, Farhad Bayat, and Adel Nadjaran Toosi. 2019. Collective hybrid intelligence: towards a conceptual framework. *International Journal of Crowd Science* 3, 2 (2019), 198–220.
- [14] Colleen Tully-Wilson, Richard Bojack, Prudence M Millea, Helen M Stallman, Andrew Allen, and Jonathan Mason. 2021. Self-perceptions of aging: A systematic review of longitudinal studies. *Psychology and Aging* 36, 7 (2021), 773.
- [15] Lawrence E Ugwu and Erhabor Sunday Idemudia. 2025. Perceptions of Ageing and Psychological Well-Being in Older Adults: A Systematic Review and Meta-Analysis. *Research on Aging* (2025), 01640275251391173.
- [16] Zezhong Wang, Suyu Hao, and Sheelagh Carpendale. 2024. Card-Based Approach to Engage Exploring Ethics in AI for Data Visualization. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. 69:1–69:7. <https://doi.org/10.1145/3613905.3650972>
- [17] Mingyue Zhu, Huimin Chen, Xiaotong Ding, and Zheng Li. 2025. Effects of self-perception of aging interventions in older adults: A systematic review and meta-analysis. *The Gerontologist* 65, 4 (2025), gnae127.